

Worked Example Solution: Convolution and Pooling Operations

Problem 1: Convolution and Max Pooling Pipeline

Problem Setup: You are given an input tensor $\mathbf{X} \in \mathbb{R}^{4 \times 4 \times 1}$ ($C_{\text{in}} = 1$) and a filter kernel $\mathbf{K} \in \mathbb{R}^{2 \times 2 \times 1}$ ($C_{\text{out}} = 1$), with **Scalar Bias** $b = 0$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{::,1} = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}, \quad \mathbf{K}_{::,1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Part A: Convolution Operation

Assuming stride $s = 1$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature map, then calculate all values of the output feature map $\mathbf{Y}$ after adding bias $b = 0$ and applying activation $\text{ReLU}(z) = \max(0, z)$.

1. Output Dimensions:

- **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{4 - 2 + 2(0)}{1} \right\rfloor + 1 = \mathbf{3}$
- **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{4 - 2 + 2(0)}{1} \right\rfloor + 1 = \mathbf{3}$
- **Channels (C):** $C_{\text{out}} = \mathbf{1}$ (1 filter applied across all input channels)
 $\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = \mathbf{3 \times 3 \times 1}$ ($\mathbf{Y} \in \mathbb{R}^{3 \times 3 \times 1}$)

2. Feature Map Calculations:

- (a) $Y_1(0, 0) = \text{ReLU}(1(1) + 1(1) + 0) = \text{ReLU}(2) = \mathbf{2}$
- (b) $Y_1(0, 1) = \text{ReLU}(0(1) + 1(1) + 0) = \text{ReLU}(1) = \mathbf{1}$
- (c) $Y_1(0, 2) = \text{ReLU}(2(1) + 0(1) + 0) = \text{ReLU}(2) = \mathbf{2}$
- (d) $Y_1(1, 0) = \text{ReLU}(0(1) + 0(1) + 0) = \text{ReLU}(0) = \mathbf{0}$
- (e) $Y_1(1, 1) = \text{ReLU}(1(1) + 1(1) + 0) = \text{ReLU}(2) = \mathbf{2}$
- (f) $Y_1(1, 2) = \text{ReLU}(1(1) + 2(1) + 0) = \text{ReLU}(3) = \mathbf{3}$
- (g) $Y_1(2, 0) = \text{ReLU}(2(1) + 1(1) + 0) = \text{ReLU}(3) = \mathbf{3}$
- (h) $Y_1(2, 1) = \text{ReLU}(0(1) + 0(1) + 0) = \text{ReLU}(0) = \mathbf{0}$
- (i) $Y_1(2, 2) = \text{ReLU}(1(1) + 1(1) + 0) = \text{ReLU}(2) = \mathbf{2}$

Resulting Matrix: $\mathbf{Y}_{::,1} = \begin{bmatrix} \mathbf{2} & \mathbf{1} & \mathbf{2} \\ \mathbf{0} & \mathbf{2} & \mathbf{3} \\ \mathbf{3} & \mathbf{0} & \mathbf{2} \end{bmatrix} \quad (\mathbf{Y} \in \mathbb{R}^{3 \times 3 \times 1})$

Part B: Max Pooling

Apply a 2×2 Max Pooling filter with stride $s = 1$ and padding $p = 0$ to the feature map $\mathbf{Y}$ you computed in Part A.

1. Output Dimensions:

- **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{3 - 2 + 2(0)}{1} \right\rfloor + 1 = \mathbf{2}$
- **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{3 - 2 + 2(0)}{1} \right\rfloor + 1 = \mathbf{2}$
- **Channels (C):** $C_{\text{out}} = \mathbf{1}$ (pooling operates per channel independently, preserving depth)
 $\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = \mathbf{2 \times 2 \times 1}$ ($\mathbf{Y} \in \mathbb{R}^{2 \times 2 \times 1}$)

2. Feature Map Calculations:

- (a) $Y_1(0, 0) = \max(2, 1, 0, 2) = \mathbf{2}$
- (b) $Y_1(0, 1) = \max(1, 2, 2, 3) = \mathbf{3}$
- (c) $Y_1(1, 0) = \max(0, 2, 3, 0) = \mathbf{3}$
- (d) $Y_1(1, 1) = \max(2, 3, 0, 2) = \mathbf{3}$

Resulting Matrix: $\mathbf{Y}_{::,1} = \begin{bmatrix} \mathbf{2} & \mathbf{3} \\ \mathbf{3} & \mathbf{3} \end{bmatrix} \quad (\mathbf{Y} \in \mathbb{R}^{2 \times 2 \times 1})$

Problem 2: Multi-Channel Convolution and Average Pooling Pipeline

Problem Setup: You are given an input tensor $\mathbf{X} \in \mathbb{R}^{3 \times 3 \times 2}$ ($C_{\text{in}} = 2$) and a filter kernel $\mathbf{K} \in \mathbb{R}^{2 \times 2 \times 2}$ ($C_{\text{out}} = 1$) with **Scalar Bias** $b = -1$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{:, :, 1} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{X}_{:, :, 2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{:, :, 1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{K}_{:, :, 2} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Part A: Multi-Channel Convolution Operation

Assuming stride $s = 1$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature map, then calculate all values of the output feature map $\mathbf{Y}$ after adding bias $b = -1$ and applying activation $\text{ReLU}(z) = \max(0, z)$.

1. Output Dimensions:

- **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{3 - 2 + 2(0)}{1} \right\rfloor + 1 = 2$
 - **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{3 - 2 + 2(0)}{1} \right\rfloor + 1 = 2$
 - **Channels (C):** $C_{\text{out}} = 1$ (1 filter applied across all input channels)
- $\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = 2 \times 2 \times 1$ ($\mathbf{Y} \in \mathbb{R}^{2 \times 2 \times 1}$)

2. Feature Map Calculations:

- (a) $Y_1(0, 0) = \text{ReLU}((\mathbf{X}_{:, :, 1} * \mathbf{K}_{:, :, 1})(0, 0) + (\mathbf{X}_{:, :, 2} * \mathbf{K}_{:, :, 2})(0, 0) + b) = \text{ReLU}(3 + 0 - 1) = \text{ReLU}(2) = 2$
- (b) $Y_1(0, 1) = \text{ReLU}((\mathbf{X}_{:, :, 1} * \mathbf{K}_{:, :, 1})(0, 1) + (\mathbf{X}_{:, :, 2} * \mathbf{K}_{:, :, 2})(0, 1) + b) = \text{ReLU}(1 + 0 - 1) = \text{ReLU}(0) = 0$
- (c) $Y_1(1, 0) = \text{ReLU}((\mathbf{X}_{:, :, 1} * \mathbf{K}_{:, :, 1})(1, 0) + (\mathbf{X}_{:, :, 2} * \mathbf{K}_{:, :, 2})(1, 0) + b) = \text{ReLU}(1 + 0 - 1) = \text{ReLU}(0) = 0$
- (d) $Y_1(1, 1) = \text{ReLU}((\mathbf{X}_{:, :, 1} * \mathbf{K}_{:, :, 1})(1, 1) + (\mathbf{X}_{:, :, 2} * \mathbf{K}_{:, :, 2})(1, 1) + b) = \text{ReLU}(3 + 0 - 1) = \text{ReLU}(2) = 2$

Resulting Matrix: $\mathbf{Y}_{:, :, 1} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ ($\mathbf{Y} \in \mathbb{R}^{2 \times 2 \times 1}$)

Part B: Average Pooling

Apply a 2×2 Average Pooling filter with stride $s = 1$ and padding $p = 0$ to the feature map $\mathbf{Y}$ you computed in Part A.

1. Output Dimensions:

- **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{2 - 2 + 2(0)}{1} \right\rfloor + 1 = 1$
 - **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{2 - 2 + 2(0)}{1} \right\rfloor + 1 = 1$
 - **Channels (C):** $C_{\text{out}} = 1$ (pooling operates per channel independently, preserving depth)
- $\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = 1 \times 1 \times 1$ ($\mathbf{Y} \in \mathbb{R}^{1 \times 1 \times 1}$)

2. Feature Map Calculations:

(a) $Y_1(0, 0) = \frac{1}{2 \times 2}(2 + 0 + 0 + 2) = \frac{4}{4} = 1$

Resulting Matrix: $\mathbf{Y}_{:, :, 1} = [1]$ ($\mathbf{Y} \in \mathbb{R}^{1 \times 1 \times 1}$)

Problem 3: Multi-Filter Convolution and Max Pooling with Rectangular Tensors

Problem Setup: You are given a rectangular input tensor $\mathbf{X} \in \mathbb{R}^{3 \times 4 \times 2}$ ($H_{\text{in}} = 3, W_{\text{in}} = 4, C_{\text{in}} = 2$) and **two independent filters** $\mathbf{K}^{(1)}, \mathbf{K}^{(2)} \in \mathbb{R}^{2 \times 2 \times 2}$ ($C_{\text{out}} = 2$) with **Biases** $b_1 = 0, b_2 = -1$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{::,1} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \end{bmatrix}, \quad \mathbf{X}_{::,2} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{K}_{::,1}^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{K}_{::,2}^{(1)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{::,1}^{(2)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{::,2}^{(2)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Part A: Multi-Filter Convolution Operation

Assuming stride $s = 1$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature tensor, then calculate all values of the output feature tensor $\mathbf{Y}$ after adding per-filter biases and applying activation $\text{ReLU}(z) = \max(0, z)$.

1. Output Dimensions:

- **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{3 - 2 + 2(0)}{1} \right\rfloor + 1 = 2$
- **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{4 - 2 + 2(0)}{1} \right\rfloor + 1 = 3$
- **Channels (C):** $C_{\text{out}} = 2$ (2 independent filters applied across all input channels)
 $\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = 2 \times 3 \times 2$ ($\mathbf{Y} \in \mathbb{R}^{2 \times 3 \times 2}$)

2. Feature Map Calculations:**Channel 1 ($\mathbf{Y}_{::,1}$, Filter $\mathbf{K}^{(1)}$, $b_1 = 0$, ReLU):****Channel 2 ($\mathbf{Y}_{::,2}$, Filter $\mathbf{K}^{(2)}$, $b_2 = -1$, ReLU):**

- | | |
|---|--|
| (a) $Y_1(0, 0) = \text{ReLU}((1+1) + (1+1) + 0) = \text{ReLU}(4) = 4$ | (g) $Y_2(0, 0) = \text{ReLU}((0+0) + (0+0) - 1) = \text{ReLU}(-1) = 0$ |
| (b) $Y_1(0, 1) = \text{ReLU}((0+1) + (0+0) + 0) = \text{ReLU}(1) = 1$ | (h) $Y_2(0, 1) = \text{ReLU}((1+1) + (1+2) - 1) = \text{ReLU}(4) = 4$ |
| (c) $Y_1(0, 2) = \text{ReLU}((1+2) + (1+2) + 0) = \text{ReLU}(6) = 6$ | (i) $Y_2(0, 2) = \text{ReLU}((0+1) + (0+0) - 1) = \text{ReLU}(0) = 0$ |
| (d) $Y_1(1, 0) = \text{ReLU}((0+0) + (0+0) + 0) = \text{ReLU}(0) = 0$ | (j) $Y_2(1, 0) = \text{ReLU}((1+1) + (1+1) - 1) = \text{ReLU}(3) = 3$ |
| (e) $Y_1(1, 1) = \text{ReLU}((1+2) + (2+1) + 0) = \text{ReLU}(6) = 6$ | (k) $Y_2(1, 1) = \text{ReLU}((1+0) + (0+1) - 1) = \text{ReLU}(1) = 1$ |
| (f) $Y_1(1, 2) = \text{ReLU}((1+1) + (0+1) + 0) = \text{ReLU}(3) = 3$ | (l) $Y_2(1, 2) = \text{ReLU}((2+2) + (2+1) - 1) = \text{ReLU}(6) = 6$ |

Resulting Matrix: $\mathbf{Y}_{::,1} = \begin{bmatrix} 4 & 1 & 6 \\ 0 & 6 & 3 \end{bmatrix}, \quad \mathbf{Y}_{::,2} = \begin{bmatrix} 0 & 4 & 0 \\ 3 & 1 & 6 \end{bmatrix}$ ($\mathbf{Y} \in \mathbb{R}^{2 \times 3 \times 2}$)

Part B: Max Pooling

Apply a 2×2 Max Pooling filter with stride $s = 1$ and padding $p = 0$ to each channel of the feature map $\mathbf{Y}$ you computed in Part A.

1. Output Dimensions:

- **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{2 - 2 + 2(0)}{1} \right\rfloor + 1 = 1$
- **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{3 - 2 + 2(0)}{1} \right\rfloor + 1 = 2$
- **Channels (C):** $C_{\text{out}} = 2$ (pooling operates per channel independently, preserving depth)
 $\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = 1 \times 2 \times 2$ ($\mathbf{Y} \in \mathbb{R}^{1 \times 2 \times 2}$)

2. Feature Map Calculations:**Channel 1 ($\mathbf{Y}_{::,1}$):****Channel 2 ($\mathbf{Y}_{::,2}$):**

- | | |
|--|--|
| (a) $Y_1(0, 0) = \max(4, 1, 0, 6) = 6$ | (c) $Y_2(0, 0) = \max(0, 4, 3, 1) = 4$ |
| (b) $Y_1(0, 1) = \max(1, 6, 6, 3) = 6$ | (d) $Y_2(0, 1) = \max(4, 0, 1, 6) = 6$ |

Resulting Matrix: $\mathbf{Y}_{::,1} = \begin{bmatrix} 6 & 6 \end{bmatrix}, \quad \mathbf{Y}_{::,2} = \begin{bmatrix} 4 & 6 \end{bmatrix}$ ($\mathbf{Y} \in \mathbb{R}^{1 \times 2 \times 2}$)

Problem 4: Convolution and Max Pooling with Stride $s = 2$

Problem Setup: You are given an input tensor $\mathbf{X} \in \mathbb{R}^{5 \times 5 \times 1}$ ($C_{\text{in}} = 1$) and a filter kernel $\mathbf{K} \in \mathbb{R}^{3 \times 3 \times 1}$ ($C_{\text{out}} = 1$) with **Scalar Bias** $b = -1$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{:, :, 1} = \begin{bmatrix} 2 & 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 & 0 \\ 1 & 0 & 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{K}_{:, :, 1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Part A: Convolution Operation with Stride $s = 2$

Assuming stride $s = 2$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature map, then calculate all values of the output feature map $\mathbf{Y}$ after adding bias $b = -1$ and applying activation $\text{ReLU}(z) = \max(0, z)$.

1. Output Dimensions:

• **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{5 - 3 + 2(0)}{2} \right\rfloor + 1 = \left\lfloor \frac{2}{2} \right\rfloor + 1 = \mathbf{2}$

• **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{5 - 3 + 2(0)}{2} \right\rfloor + 1 = \left\lfloor \frac{2}{2} \right\rfloor + 1 = \mathbf{2}$

• **Channels (C):** $C_{\text{out}} = \mathbf{1}$ (1 filter applied across all input channels)

$\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = \mathbf{2 \times 2 \times 1}$ ($\mathbf{Y} \in \mathbb{R}^{2 \times 2 \times 1}$)

2. Feature Map Calculations:

(a) $Y_1(0, 0) = \text{ReLU}([2 - 1 + 1 - 1 + 2] - 1) = \text{ReLU}(2) = \mathbf{2}$ (top-left: rows 0..2, cols 0..2)

(b) $Y_1(0, 1) = \text{ReLU}([1 - 3 + 2 - 2 + 1] - 1) = \text{ReLU}(-2) = \mathbf{0}$ (top-right: rows 0..2, cols 2..4)

(c) $Y_1(1, 0) = \text{ReLU}([1 - 2 + 0 - 1 + 0] - 1) = \text{ReLU}(-3) = \mathbf{0}$ (bottom-left: rows 2..4, cols 0..2)

(d) $Y_1(1, 1) = \text{ReLU}([2 - 1 + 3 - 0 + 2] - 1) = \text{ReLU}(5) = \mathbf{5}$ (bottom-right: rows 2..4, cols 2..4)

Resulting Matrix: $\mathbf{Y}_{:, :, 1} = \begin{bmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{5} \end{bmatrix}$ ($\mathbf{Y} \in \mathbb{R}^{2 \times 2 \times 1}$)

Part B: Max Pooling with Stride $s = 2$

Apply a 2×2 Max Pooling filter with stride $s = 2$ and padding $p = 0$ to the feature map $\mathbf{Y}$ you computed in Part A.

1. Output Dimensions:

• **Height (H):** $H_{\text{out}} = \left\lfloor \frac{H - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{2 - 2 + 2(0)}{2} \right\rfloor + 1 = \left\lfloor \frac{0}{2} \right\rfloor + 1 = \mathbf{1}$

• **Width (W):** $W_{\text{out}} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{2 - 2 + 2(0)}{2} \right\rfloor + 1 = \left\lfloor \frac{0}{2} \right\rfloor + 1 = \mathbf{1}$

• **Channels (C):** $C_{\text{out}} = \mathbf{1}$ (pooling operates per channel independently, preserving depth)

$\Rightarrow H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}} = \mathbf{1 \times 1 \times 1}$ ($\mathbf{Y} \in \mathbb{R}^{1 \times 1 \times 1}$)

2. Feature Map Calculations:

(a) $Y_1(0, 0) = \max(2, 0, 0, 5) = \mathbf{5}$

Resulting Matrix: $\mathbf{Y}_{:, :, 1} = [\mathbf{5}]$ ($\mathbf{Y} \in \mathbb{R}^{1 \times 1 \times 1}$)